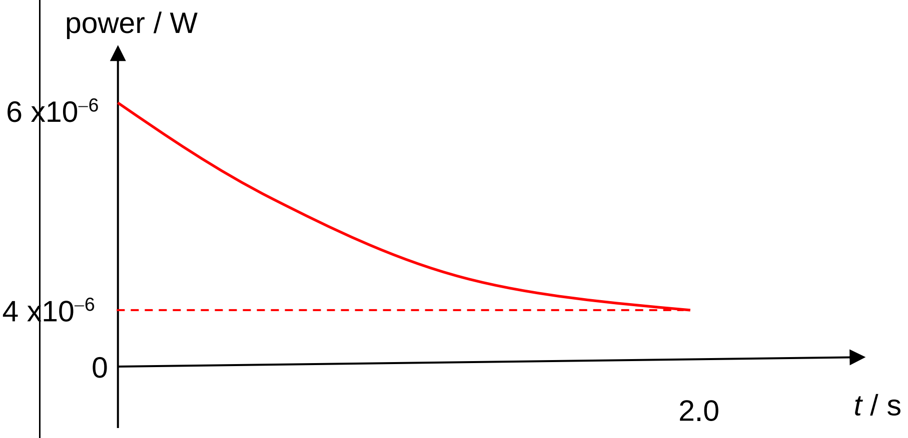
10	**(a)**	**(i)**	A wave in which <u>**energy is carried from one point to another**</u> by means of vibrations or oscillations within the wave. The <u>displacements of the particles in the wave are along the direction of transfer of energy</u> of the wave.	B1 B1
	(a)	**(ii)**	Particle Y experiences <u>lowest/minimum</u> pressure since <u>its neighbouring particles move away from</u> it simultaneously at that instant.	B1 B1
	(a)	**(iii)**	Distance of 2 wavelength between X and Z wavelength λ = 5.1 / 2 = 2.55 m Since v = f λ = 130 x 2.55 = 330 m s^{-1}	 M1 A1
	(a)	**(iv)**	phase angle $= (\frac{1.5\lambda}{\lambda})2\pi$ $= 9.43\ \text{rad}$ (accept π or 3π)	A1
	(b)	**(i)**	Intensity of wave at position of microphone = $P / 4\pi x^2$ = $0.82/[4\pi(4.8)^2]$ = 2.83×10^{-3} W m^{-2} Assume the area of the sphere receiving the waves is πr^2, Power picked up by microphone = $I \times A$ = $(2.83 \times 10^{-3}) \times [\pi(0.025)^2]$ = 5.56×10^{-6} W ~ 6×10^{-6} W.	 M1 C1 A1

	(b)	(ii)		

(b) **(ii)**



B1

B1

Since intensity is proportional to $1/r^2$, $I = k/r^2$

When $I = 2.83 \times 10^{-3}$ W m^{-2} and $r = 4.8$ m,
$k = (2.83 \times 10^{-3})(4.8^2) = 0.0652$

Since $I = P/A$ and $I = k/r^2$, and $r = 4.8 + vt$

$P/A = k/r^2 = k/(4.8 + vt)^2$
$P = kA/(4.8 + vt)^2$

Since $k = 0.0652$, $A = \pi(0.025)^2$, $v = 0.40$, $t = 2.0$,

$P = (0.0652)(\pi(0.025)^2)/(4.8 + (0.40 \times 2.0))^2$
$= 4.08 \times 10^{-6}$
$\sim 4 \times 10^{-6}$ B1

1 mark for correct shape of graph
1 mark for labelling of points on the graph
1 mark for correct computation of power at t = 2 s.

	(c)	**(i)**	path difference = 2 x 0.20m = 0.40 m		A1
	(c)	**(ii)**	For next constructive interference to take place, path difference = nλ where n = 1. λ = 0.40m v = f λ 320 = f x 0.40 = 800 Hz		C1 A1
	(c)	**(iii)**	The sound waves from path APB and AQB are <u>channelled by the tube</u> (contextualised response) meeting in the <u>opposite directions</u>. Since both waves are of <u>equal amplitude, frequency and speed</u>, they <u>superpose and interfere</u> to form a stationary wave.		B1 B1
	(c)	**(iv)**	A.	Replace microphone with microwave detector connected to C.R.O.	B1
			B.	Accept any other reasonable answers. Since frequency of microwave is higher, the wavelength will be shorter. The path difference will be smaller as well so the distances between adjacent nodes and antinodes of the signal will be smaller.	B1